Leeson 3, 4: Supervised Machine Learning: Regression, Classification and Model Evaluation

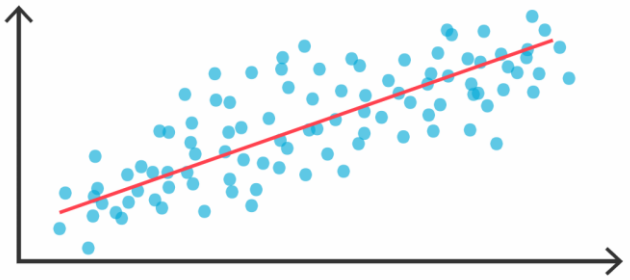
Learning objectives

- Distinguish regression from classification and identify appropriate climate applications.
- Explain how linear regression and logistic regression make predictions.
- Train a basic supervised learning model using Python and scikit-learn.
- Evaluate regression predictions using MAE and RMSE.
- Evaluate classification predictions using a confusion matrix, precision, recall and accuracy.
- Explain why baselines, chronological testing and data leakage prevention are essential for climate forecasting.

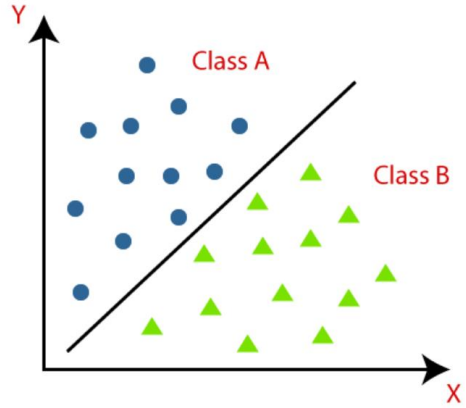
Recap

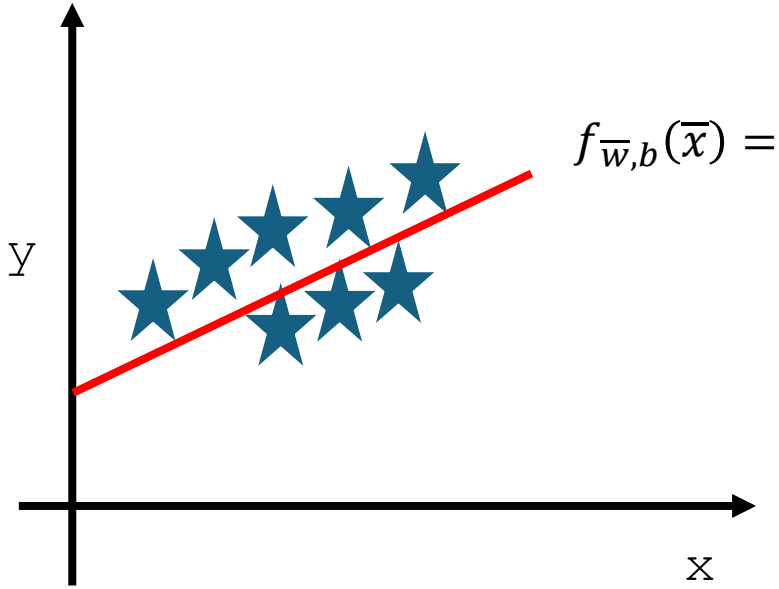
Supervised Learning

Regression



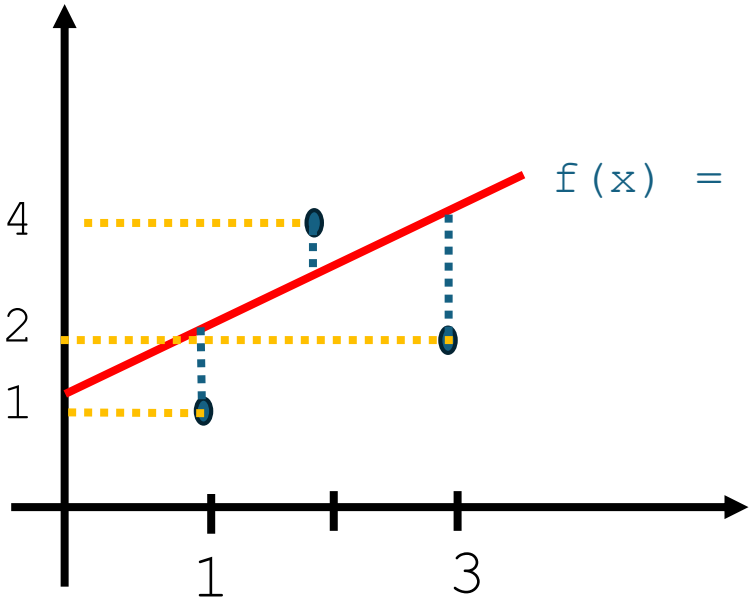
Classification





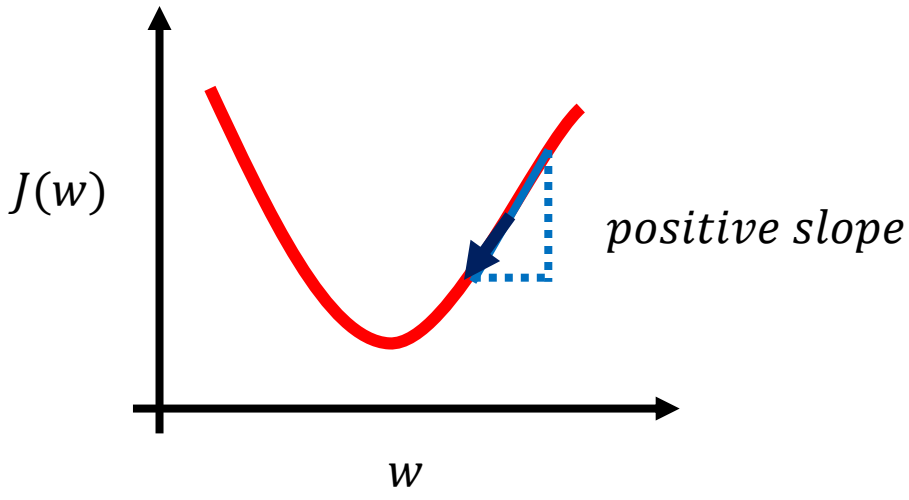
$$f_{\overline{w},b}(\overline{x}) = wx + b$$

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(\hat{y}^{(i)} - y^{(i)})^2$$

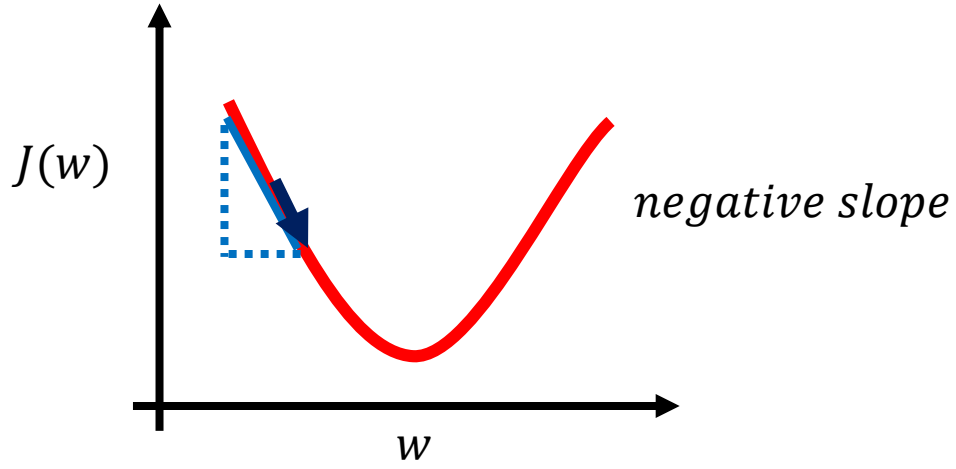



```
f(x) = 0.5*x + 1
```

$$J(w,b) = \frac{1}{2*3} * ((1.5-1)^2 + (4-2)^2 + (2.-2.5)^2)$$



$$w_j = w_j \; -\propto \frac{\partial}{\partial w_j} \; J(\overline{w}, b)$$



- **Features (X):** information supplied to the model.
- **Target (y):** the quantity or category to predict.
- **Model:** the mathematical function or structure used to generate predictions.
- **Training:** the process of fitting the model using examples with known targets.

Two major supervised learning tasks:

Regression	Classification
Predicts a numerical quantity	Predicts a category
Example: rainfall amount in mm	Example: wet day or dry day

One row represents one observation at a weather station on a particular date.
Illustrative dataset structure:

Date	Rain today (mm)	Max temp (°C)	Humidity (%)	Rain tomorrow (mm)
Day 1	0.0	27	42	0.0
Day 2	0.0	29	38	2.5
Day 3	2.5	25	61	7.0
Day 4	7.0	22	78	0.0

Terminology

- **Observation:** one station-date record.
- **Feature:** a predictor used as input.
- **Target:** the output we want to predict.
- **Prediction:** the model's estimated output.
- **Residual:** actual value minus predicted value in regression.

Regression

Climate question	Target	ML task
How much rain will fall tomorrow?	6.4 mm	Regression
What will tomorrow's maximum temperature be?	31.2 °C	Regression
Will tomorrow meet the wet-day definition?	Wet / dry	Binary classification
Which land-cover class appears in a satellite image?	Water / soil / vegetation / built-up	Multiclass classification
Will an agricultural area experience a defined risk category?	Low / medium / high	Multiclass classification or an ordinal modelling approach

Key distinction: A numerical code does not automatically make a target a regression problem. If 0, 1 and 2 represent categories, classification is generally appropriate.

Linear regression: predicting a numerical quantity

What is linear regression?
Linear regression estimates a numerical target using a linear combination of input features.

For one feature:

$$\hat{y} = b_0 + b_1 x$$

Where:

- x: input feature.
- $\hat{y}$: predicted value.
- b_0: intercept.
- b_1: coefficient or slope.

The model learns the coefficient and intercept from training data.

Multiple linear regression: using several features

Climate variables often provide complementary information. A model can use multiple features rather than relying on a single predictor.

The general equation is:

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n x_n$$

For example:

$$\widehat{Rain}_{tomorrow} = b_0 + b_1 Rain_{today} + b_2 Humidity_{today} + b_3 Temperature_{today}$$

Important concepts

- More relevant features can provide useful information.
- More features do not guarantee better predictions.
- Strong correlations between features can make individual coefficients harder to interpret.
- Relationships may be nonlinear, seasonal or location-

Multiple Feature

X1	X2	X3	X4	Y
Size(feet^2)	**Number of bedrooms**	**Number of floors**	**Age of home (years)**	**Price**
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

Dataset

Number of feature

multivariate linear regression

$$f_{\overline{w},b}(\overline{x}) = x_1 w_1 + x_2 w_2 + \cdots x_n w_n + b$$

$$\overline{w} = [w_1 \quad w_2 \quad w_3 \ldots w_n]$$

$$\overline{X} = [X_1 \quad X_2 \quad X_3 \ldots X_n]$$

$$x_1 w_1 + x_2 w_2 + \cdots x_n w_n + b$$

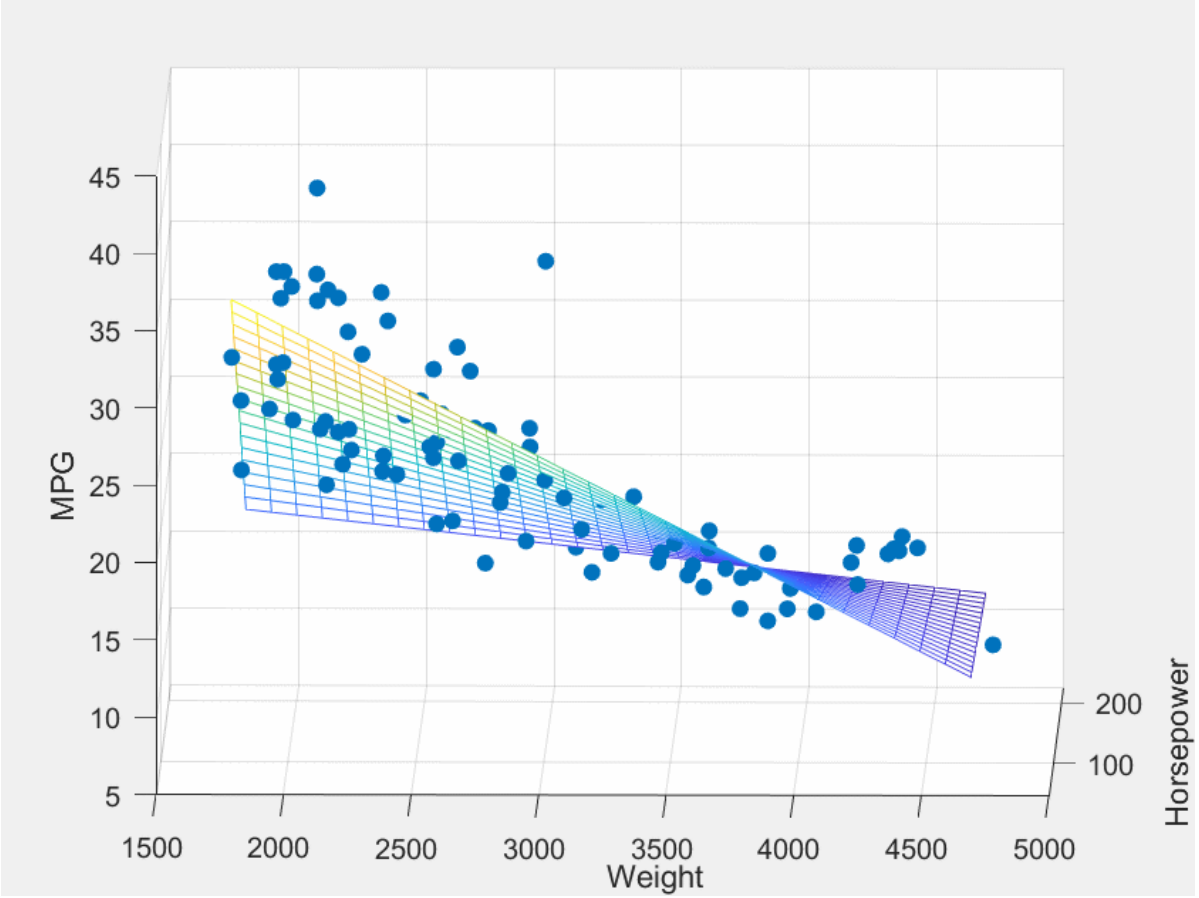
$$f_{\overline{w},b}(\overline{x}) = \overline{w} \cdot \overline{x} + b$$

$$X^{(2)} = [1416 \quad 3 \quad 2 \quad 40]$$

$$f_{\overline{w},b}(\overline{x}) = 0.1x_1 + 4x_2 + 10x_3 - 2x_4 + b \ (80)$$

size # bedroom # floors years

Multiple Feature



MPG = b0 + b1 * horsepower + b2 * weight

use two input features instead of one, the prediction space becomes **three-dimensional**

Vectorization

$$\overline{w} = [w_1 \quad w_2 \quad w_3]$$
$$\overline{x} = [x_1 \quad x_2 \quad x_3]$$

```
import numpy as np
w = np.array([1.0, -2.0, 3.2])
b = 5
x = np.array([10, 15, 20])
```

$$f_{\overline{w},b}(\overline{x}) = \sum_{j=1}^{n} x_j w_j + b$$

```
f = 0
for j in range(0,n):
    f = f + w[j]*x[j]
f = f + b
```

WITHOUT VECTORIZATION

$$f_{\overline{w},b}(\overline{x}) = x_1 w_1 + x_2 w_2 + x_3 w_3 + b$$

```
f = w[0]*x[0] +
    w[1]*x[1] +
    w[2]*x[2] + b
```

VECTORIZATION

$$f_{\overline{w},b}(\overline{x}) = \overline{x} \ \ \overline{w} + b$$

```
f = np.dot(w,x) + b
```

Gradient Descent

$$\overline{w} = [w_1 \quad w_2 \quad \ldots w_{16}]$$
$$\overline{d} = [d_1 \quad d_2 \quad \ldots d_{16}]$$

```
w = np.array([1.0, -2.0, ... 3.2])
d = np.array([0.3, 0.7, ... 0.2])
```

WITHOUT VECTORIZATION

$$w_1 = w_1 - 0.1d_1$$
$$w_2 = w_2 - 0.1d_2$$
$$\ldots$$
$$w_{16} = w_{16} - 0.1d_{16}$$

```
for j in range(0,16):
    f = w[j] - 0.1*d[j]
```

VECTORIZATION

$$\overline{w} = \overline{w} - 0.1d_1$$

```
w = w - 0.1*d
```

One Feature

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)}) * x^{(i)}$$

$$\frac{\partial}{\partial w_j} J(\overline{w}, b)$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)})$$

Simultaneously update w,b

More Than One Feature

$$w_1 = w_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)}) * x^{(i)}$$

...

$$\frac{\partial}{\partial w_j} J(\overline{w}, b)$$

$$w_n = w_n - \alpha \frac{1}{m} \sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)}) * x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)})$$

Simultaneously update w (for j = 1, 2,

Question:
You are working on a regression problem with multiple input variables to predict house prices. You have gathered data on the following features for each house: square footage, number of bedrooms, number of bathrooms, and distance from the city center. You want to build a linear regression model for this task.

Which of the following statements about the assumptions of linear regression is correct in this context?

A) There are no assumptions for linear regression when dealing with multiple input variables.
B) The assumption of multicollinearity is not relevant since there is only one dependent variable.
C) It is important to check for multicollinearity among the independent variables.
D) Assumption of linearity only applies to single input variable regression, not multiple input variable regression.

Alternative optimization method to gradient descent



Normal Equation

- Only for linear regression
- Solve for w, b without iteration

$$\theta = (X^\intercal X)^{-1} X^\intercal y$$

Feature Scaling

$$price = x_1 w_1 + x_2 w_2 + b$$

size # bedrooms

$x_1 = 2000$ $x_2 = 5$ price $= 500$k

$w_1 = 50$ $w_2 = 0.1$ $b = 50$

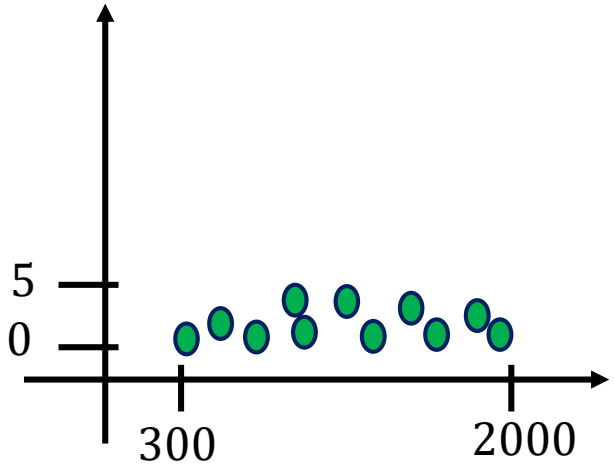
$price = 50 * 2000 + 0.1 * 5 + 50$

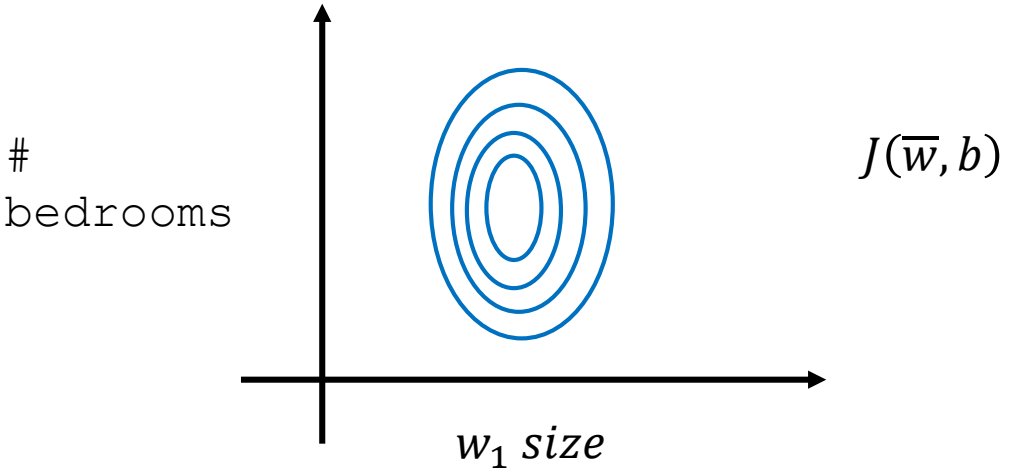
$price = 100{,}050.5$ k

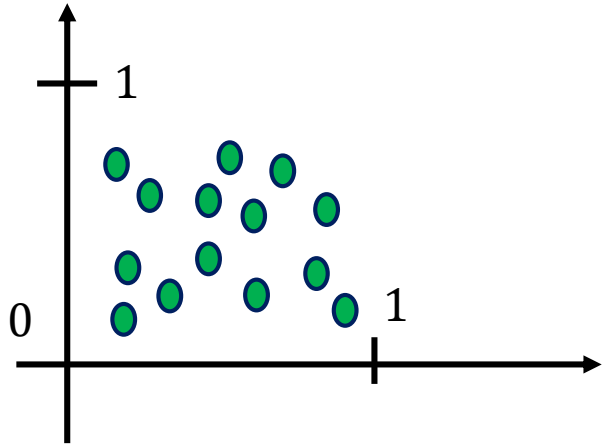
$w_1 = 0.1$ $w_2 = 50$ $b = 50$

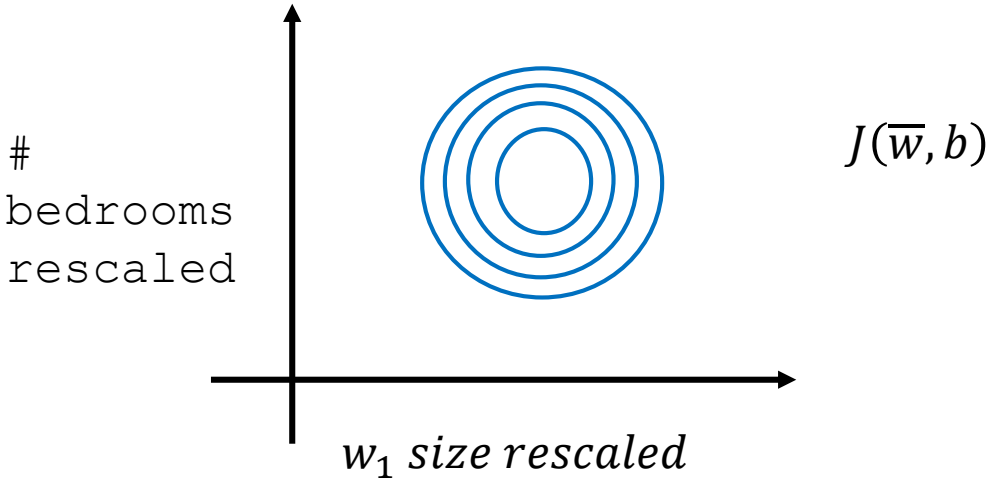
$price = 0.1 * 2000 + 0.1 * 50 + 50$

$price = 500$ k

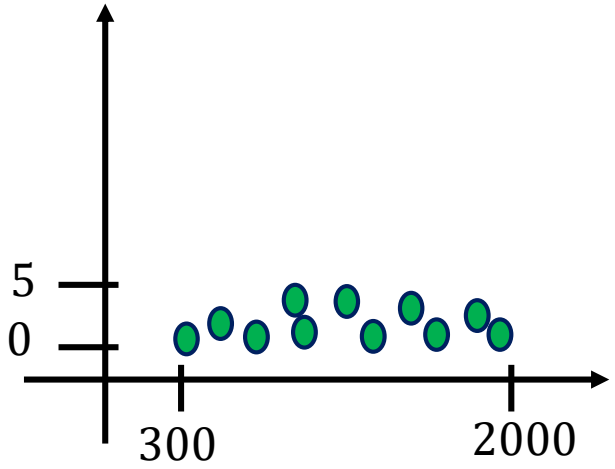

5
0
300
2000


bedrooms
$J(\overline{w}, b)$
w_1 size

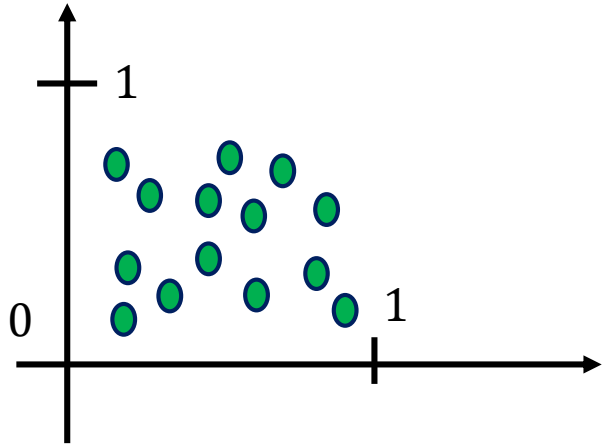

1
0
1


bedrooms rescaled
$J(\overline{w}, b)$
w_1 size rescaled

Feature Scaling



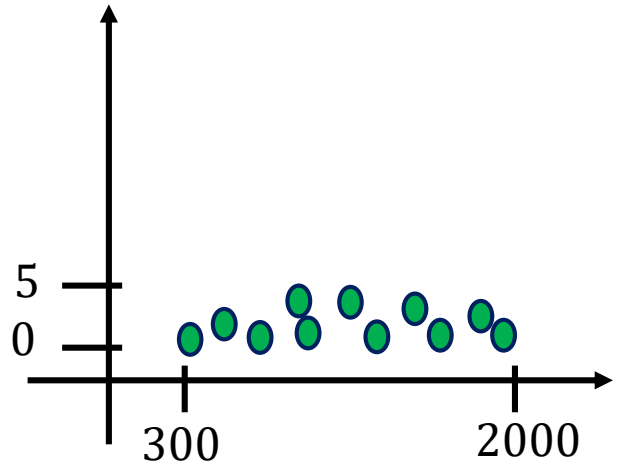
$300 < x_1 < 2000$ $\quad$ $0 < x_2 < 5$



$$x_{1,scaled} = \frac{x_1}{max} = \frac{x_1}{2000} \qquad x_2 = \frac{x_2}{max} = \frac{x_1}{5}$$

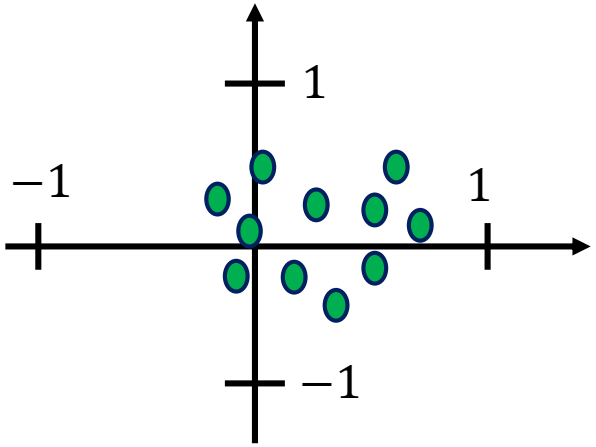
$0.15 < x_1 < 1$ $\quad$ $0 < x_2 < 1$

Mean Normalization



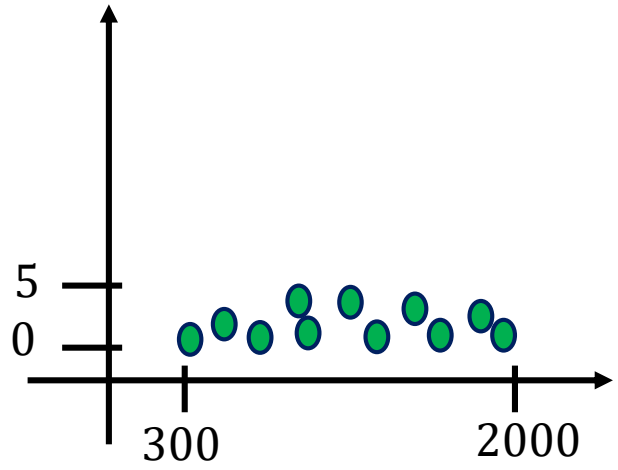
$$300 < x_1 < 2000 \qquad 0 < x_2 < 5$$

$$x_1 = \frac{(x_1 - \mu_1)}{\max - min} = \frac{(x_1 - 600)}{2000 - 300} \qquad x_2 = \frac{(x_2 - \mu_2)}{\max - min} = \frac{(x_1 - 2.3)}{5 - 0}$$



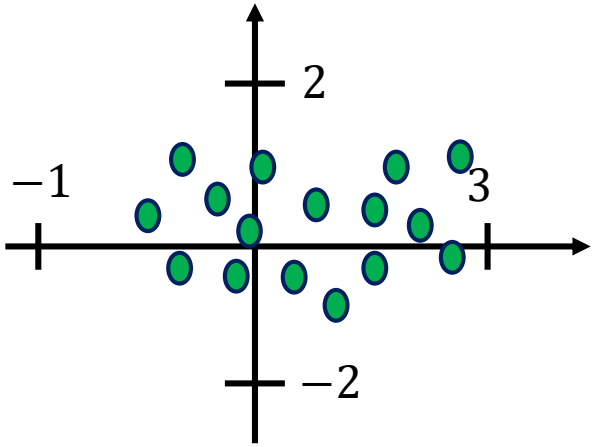
$$-0.18 < x_1 < 0.82 \qquad -0.46 < x_2 < 0.54$$

Z Score Normalization



$$300 < x_1 < 2000 \qquad 0 < x_2 < 5$$

$$x_1 = \frac{(x_1 - \mu_1)}{\sigma_1} = \frac{(x_1 - 600)}{450} \qquad x_2 = \frac{(x_2 - \mu_2)}{\sigma_2} = \frac{(x_1 - 2.3)}{1.4}$$



$$-0.67 < x_1 < 3.1 \qquad -1.6 < x_2 < 1.9$$

Feature Scaling

$$-1 \leq x_j \leq 1$$
$$-3 \leq x_j \leq 3$$
$$-0.3 \leq x_j \leq 0.3$$

Acceptable Range

$0 \leq x_j \leq 3$ Acceptable

$-2 \leq x_j \leq 0.5$ Acceptable

$-100 \leq x_j \leq 100$ Too large

$-0.001 \leq x_j \leq 0.001$ Too small

Question:
You're working on a machine learning project that involves a dataset with features of varying scales. The algorithm you plan to use is sensitive to the scale of input features. Which of the following statements about feature scaling is true?

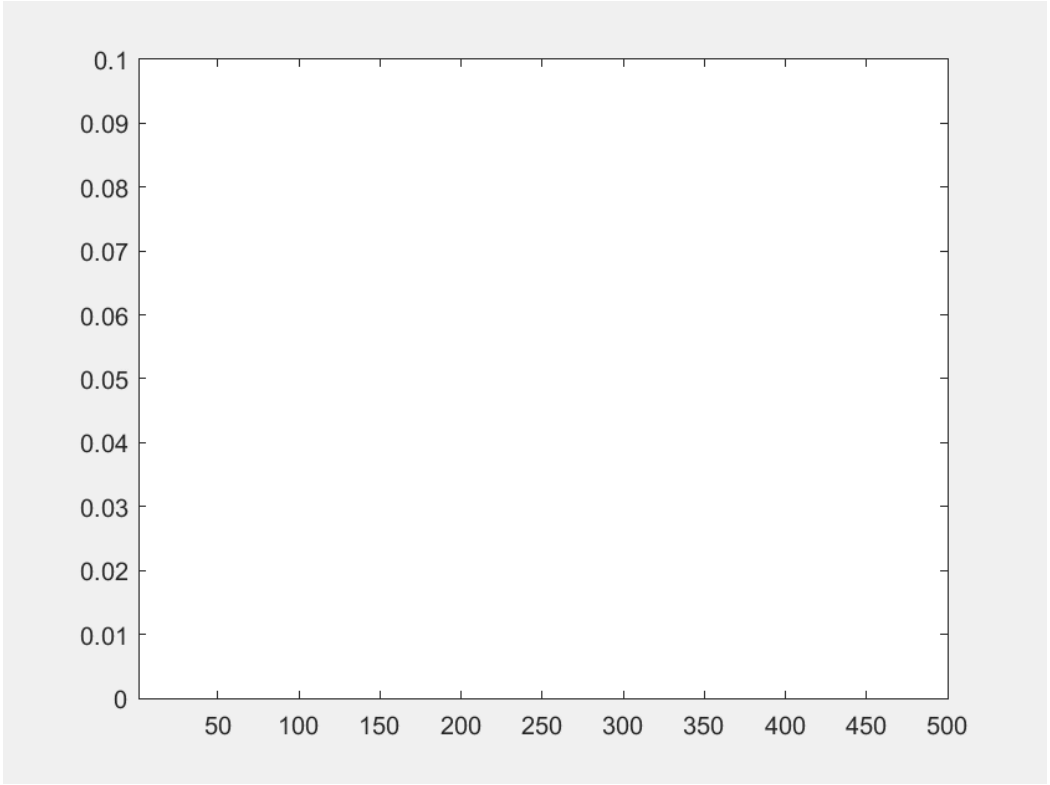
A) Feature scaling is only necessary for categorical features.
B) Feature scaling helps improve the accuracy of algorithms that are not affected by feature scales.
C) Feature scaling is primarily important for classification tasks, not regression tasks.
D) Feature scaling is important for algorithms like gradient descent-based optimization.

How can we understand gradient descent is working correctly

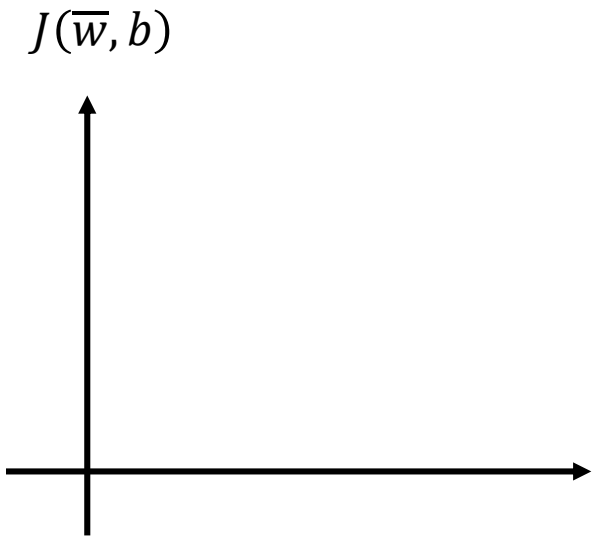


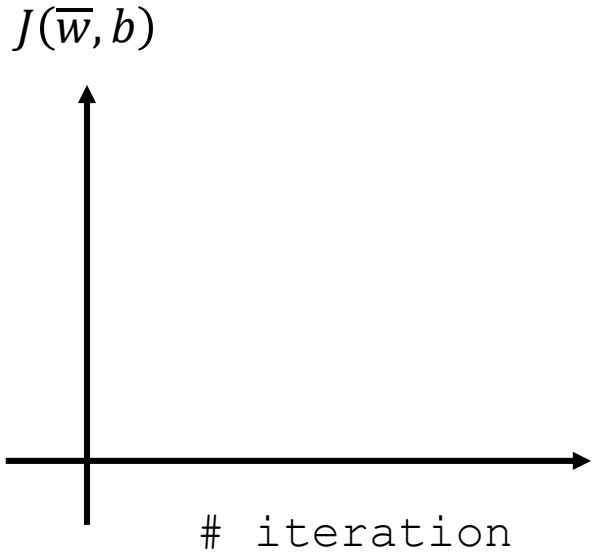
1. **Mathematical Accuracy**
2. **Convergence Behavior**
3. **Learning Rate**
4. **Gradient Checks**
5. **Plot and Visualizations**

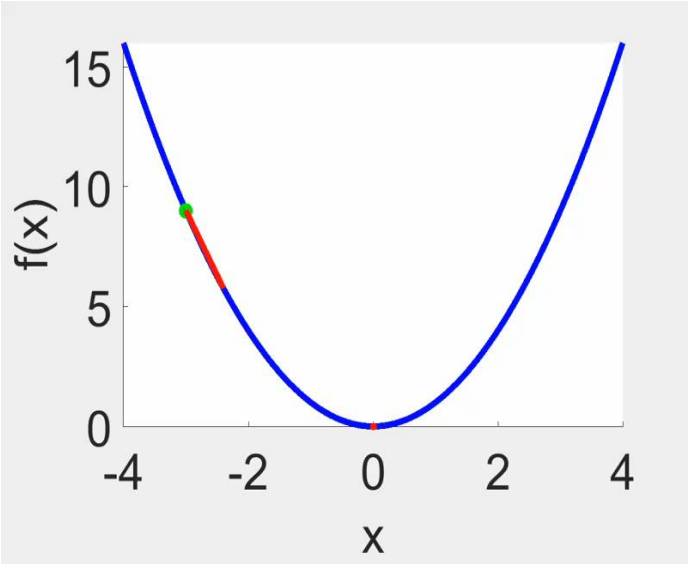
$J(\overline{w}, b)$

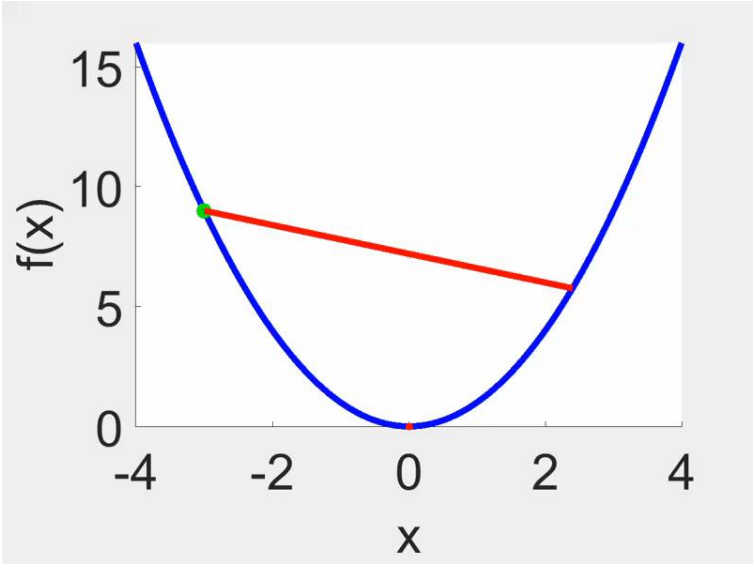


iteration







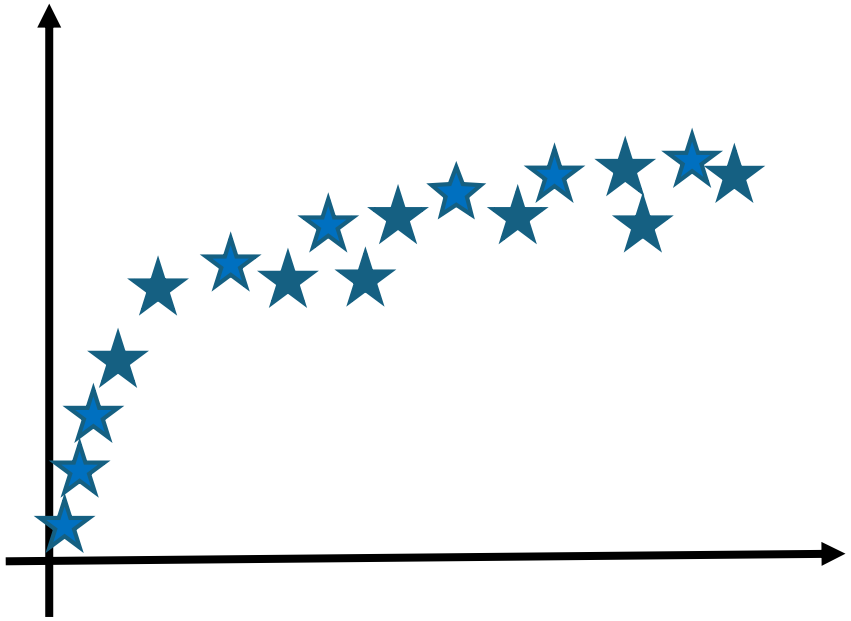


Proper learning rate

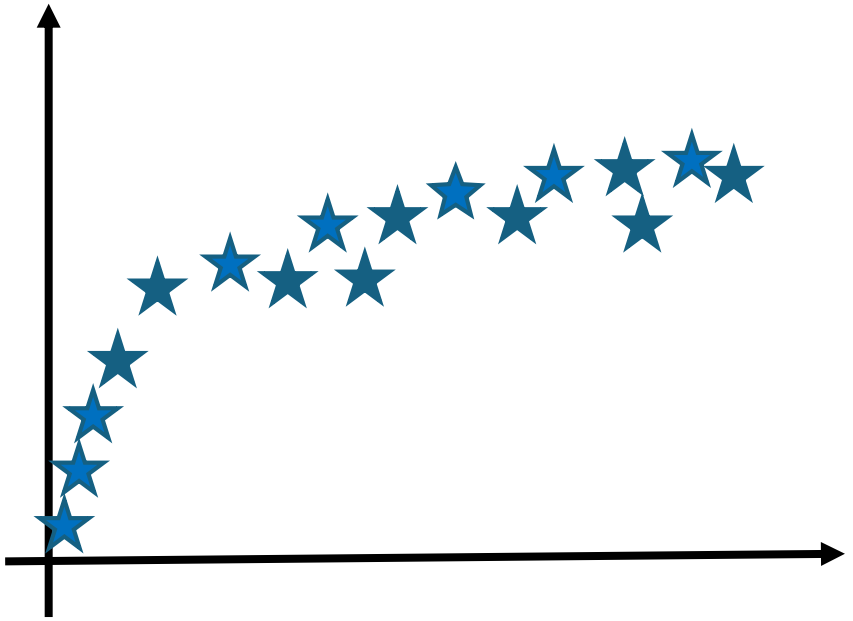
Big learning rate

If learning rate is too small, gradient descent takes too much time to converge

Polynomial Regression



$$f_{\overline{w},b}(\overline{x}) = x_1 w_1 + x_2^2 w_2 + b$$



$$f_{\overline{w},b}(\overline{x}) = x_1 w_1 + x_2^2 w_2 + x_2^3 w_2 + \ b$$

How does regression learn?

Training means finding model parameters that minimise a defined loss or cost function.

For each training example:

1. The model receives input features.
2. It calculates a predicted value.
3. The prediction is compared with the observed target.
4. The learning algorithm adjusts model parameters to reduce the
5. iterion is met.

For a predicted value $\hat{y}_i$ and actual value y_i, the residual is:

$$e_i = y_i - \hat{y}_i$$

One common training objective for linear regression is Mean Squared Error:

$$J = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

where m is the number of training examples.

Evaluating regression: MAE an RMSE

A model must be evaluated on data it did not use to fit its parameters.
Consider three held-out observations:

Actual rainfall	Predicted rainfall	Absolute error
0 mm	1 mm	1 mm
5 mm	3 mm	2 mm
12 mm	9 mm	3 mm

How to interpret the metrics

- Both metrics are expressed in the target's units.
- Lower scores indicate smaller errors on the same evaluation dataset.
- RMSE penalises large errors more strongly than MAE.

Mean Absolute Error (MAE)

$$MAE = \frac{1}{m}\sum_{i=1}^{m}|y_i - \hat{y}_i|$$

For these three observations:

$$MAE = \frac{1+2+3}{3} = 2\,\text{mm}$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}$$

For these observations:

$$RMSE = \sqrt{\frac{1+4+9}{3}} \approx 2.16\,\text{mm}$$

Always compare against a baseline

How do we know machine learning actually improves a prediction?
A baseline is a simple, transparent prediction method used as a point of comparisor
For rainfall regression, possible baselines include:
Predicting zero rainfall for every observation.
Predicting an appropriate historical seasonal average.
Using a persistence or climatology baseline, where appropriate to the task.
For classification, a baseline could predict the most common class in the training
Evaluation procedure
Define the baseline before comparing models.
Generate baseline predictions for the same held-out dates.
Generate ML predictions for those dates.
Calculate the same metric for both.
Report whether the ML model improves on the baseline.

Classification

Classification predicts a category rather than an unrestricted numerical quantity.

In binary classification, the target has two classes:

- Class 0: negative class.
- Class 1: positive class.

For our rainfall example, define a wet day as a day with rainfall of at least 1.0 mm:

$$y = \begin{cases} 1 & \text{if rainfall} \geq 1.0 \text{ mm} \\ 0 & \text{if rainfall} < 1.0 \text{ mm} \end{cases}$$

Multiclass classification: Predicts one of more than two classes, such as low, medi

Logistic regression: estimating class probabilities

Despite its name, logistic regression is commonly used for classification.

It calculates a linear score from input features:

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

The score is converted into a value between 0 and 1 using the sigmoid function:

$$p(y = 1 \mid x) = \frac{1}{1 + e^{-z}}$$

Where:

- x: input feature vector.
- w: learned coefficients.
- b: intercept.
- $p(y = 1 \mid x)$: estimated probability of the positive class, given the inputs and the fitted model.

- Example
- The model outputs:
- $$ p(\text{wet day}\mid x)=0.78 $$
- Interpretation: the fitted model assigns a 78% estimated probability to the wet-day class for this observation.
- Important: The estimated probability is not a guarantee. Its reliability depends on the model, training data and whether the model is appropriately calibrated for the deployment setting.

From a probability to a class

The decision threshold determines how probabilities are converted into predicted cl

What i

It is

which the model changes its predicted class

according to the selected threshold.

At a threshold of 0.50:

For logistic regression with a threshold of 0.50, the boundary is defined by:

$$w^\top x + b = 0$$

For two numerical features, this boundary is a line in the ordinary linear logistic-regression model.

0.15	Class 0
0.42	Class 0
0.50	Class 1
0.78	Class 1
0.93	Class 1

Why the decision threshold matters

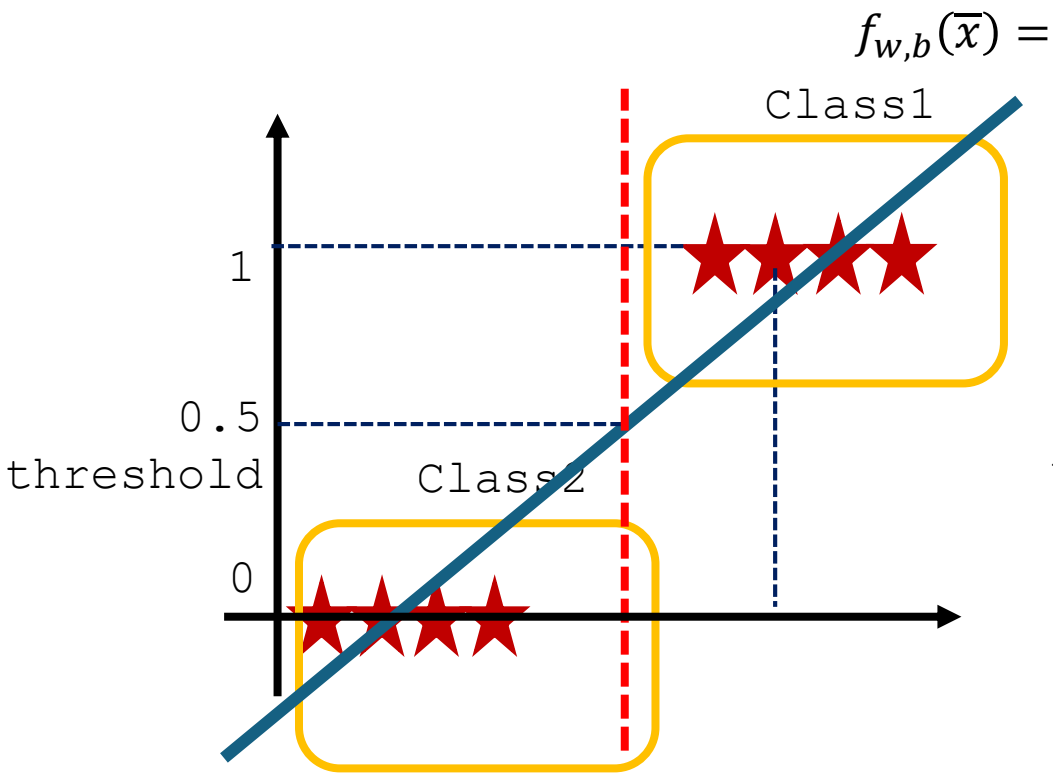
A threshold of 0.50 is a starting point, not a universal optimum.
Consider a model identifying potential drought-stress events.

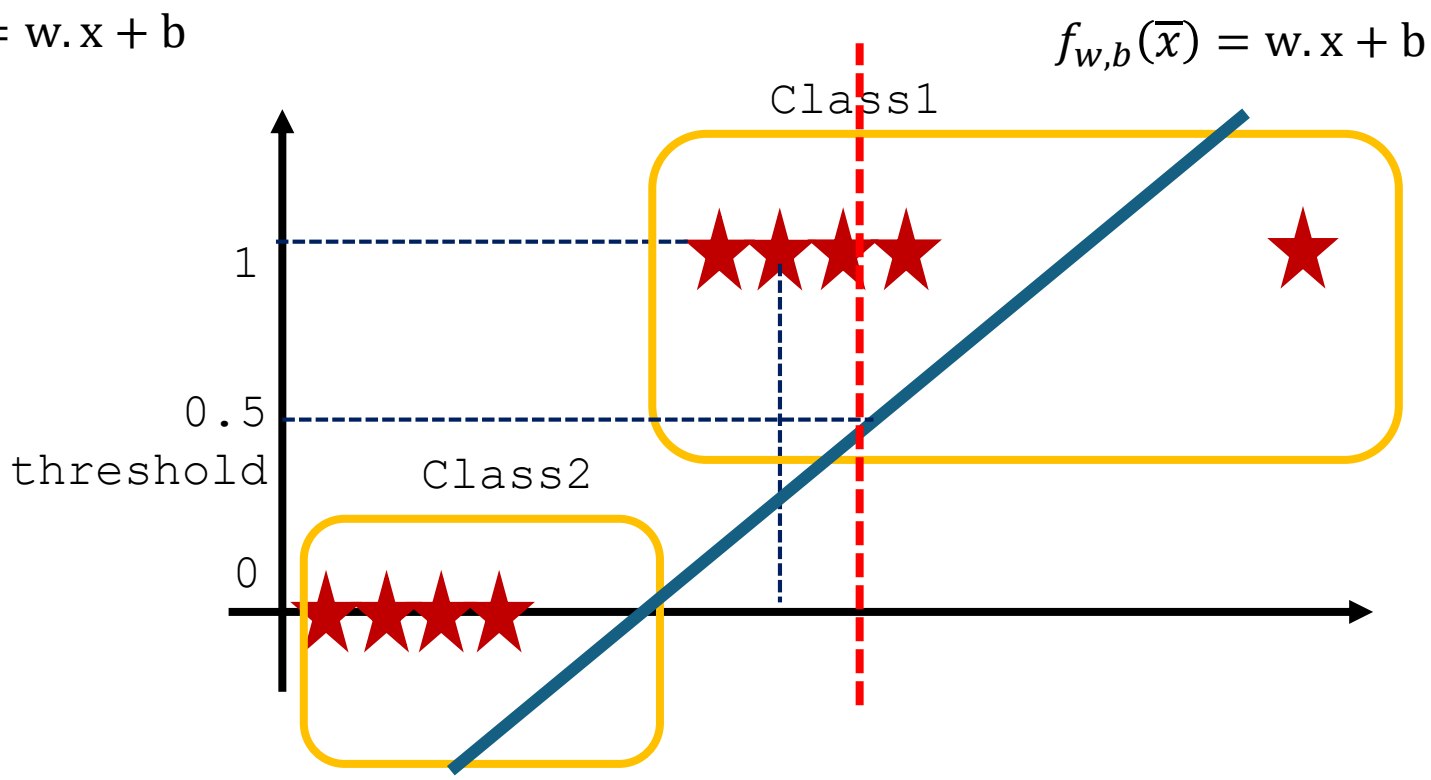
- Lower threshold: More observations may be flagged as positive. This can increase recall, but it may also increase false positives.
- Higher threshold: Fewer observations may be flagged. This may reduce false positives but can increase missed events.

Select the threshold using:

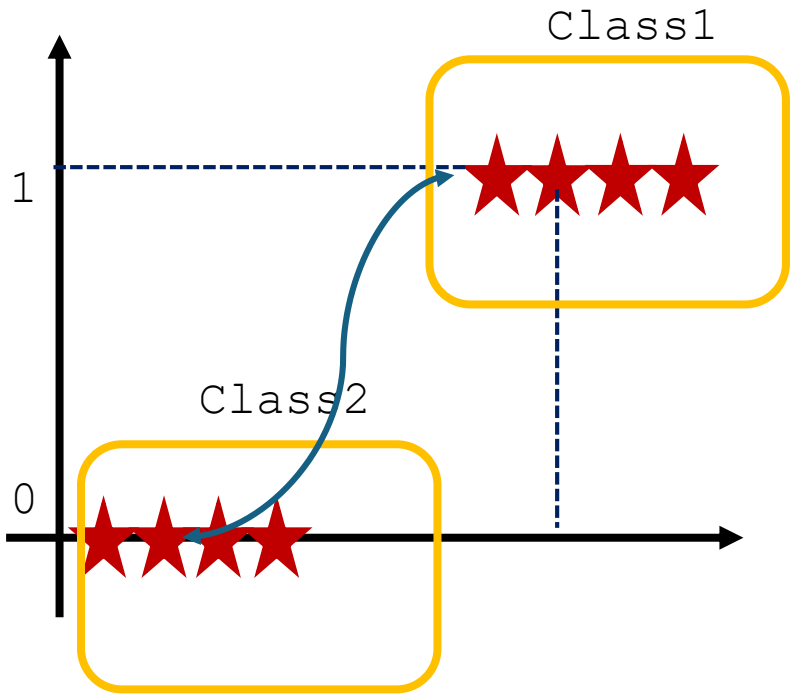
- Validation data not used for final testing.
- The consequences of false positives and false negatives.
- The desired precision and recall.
- Any operational constraints on how many alerts users can investigate.

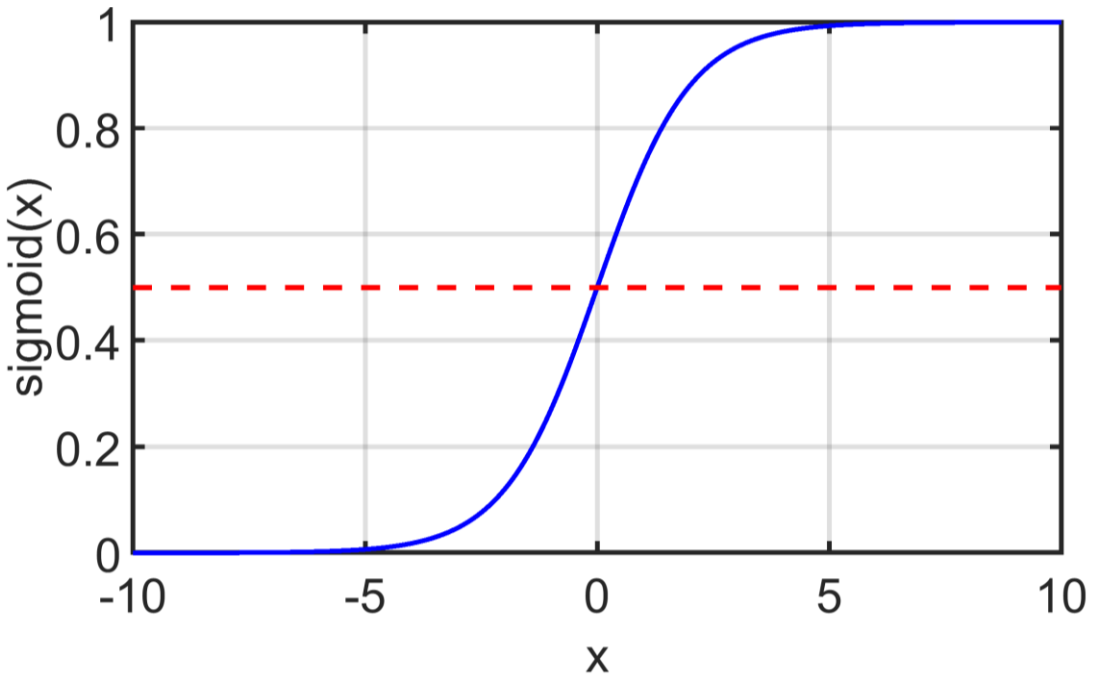
Classification problem





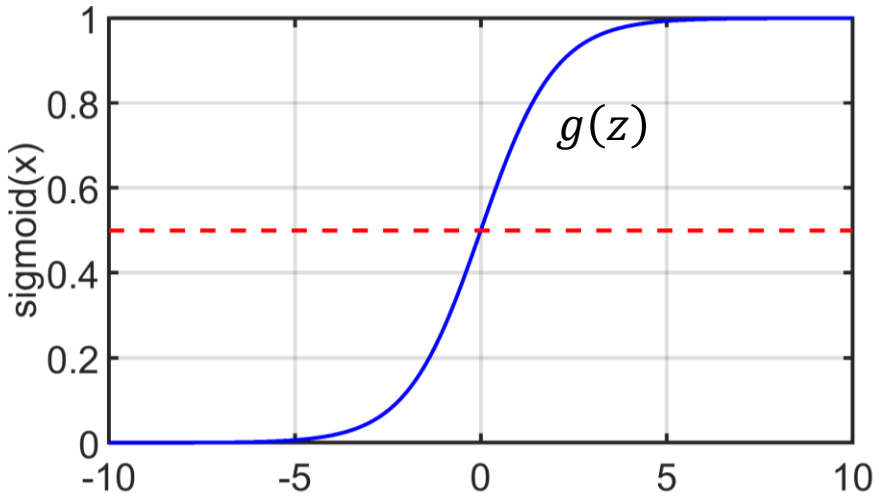
Classification problem





$$g(z) = \frac{1}{1 + e^{-z}}$$

Decision Boundary



$$g(z) = \frac{1}{1 + e^{-z}}$$

$$z = \overline{w}.\overline{x} + b$$

$Class\ 1\ when\ y = 1$

$Class\ 2\ when\ y = 0$

$$p(y = 1|x;\Theta) + p(y = 0|x;\Theta) = 1$$

$$p(y = 1) = \frac{1}{1+e^{-\Theta^T X}}$$

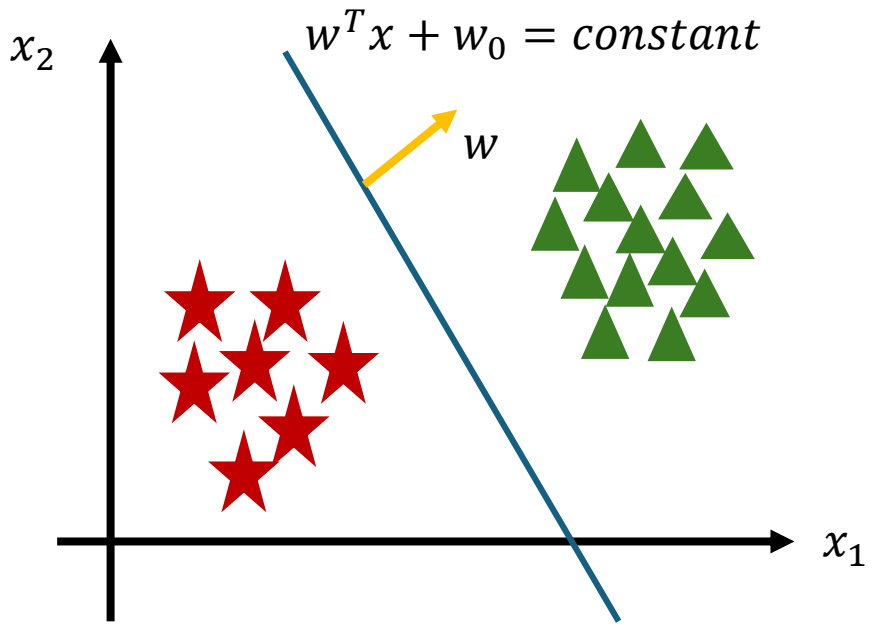
$f_{\overline{w},b}(\overline{x}) = 0.7$?

$f_{\overline{w},b}(\overline{x}) = 0.3$?

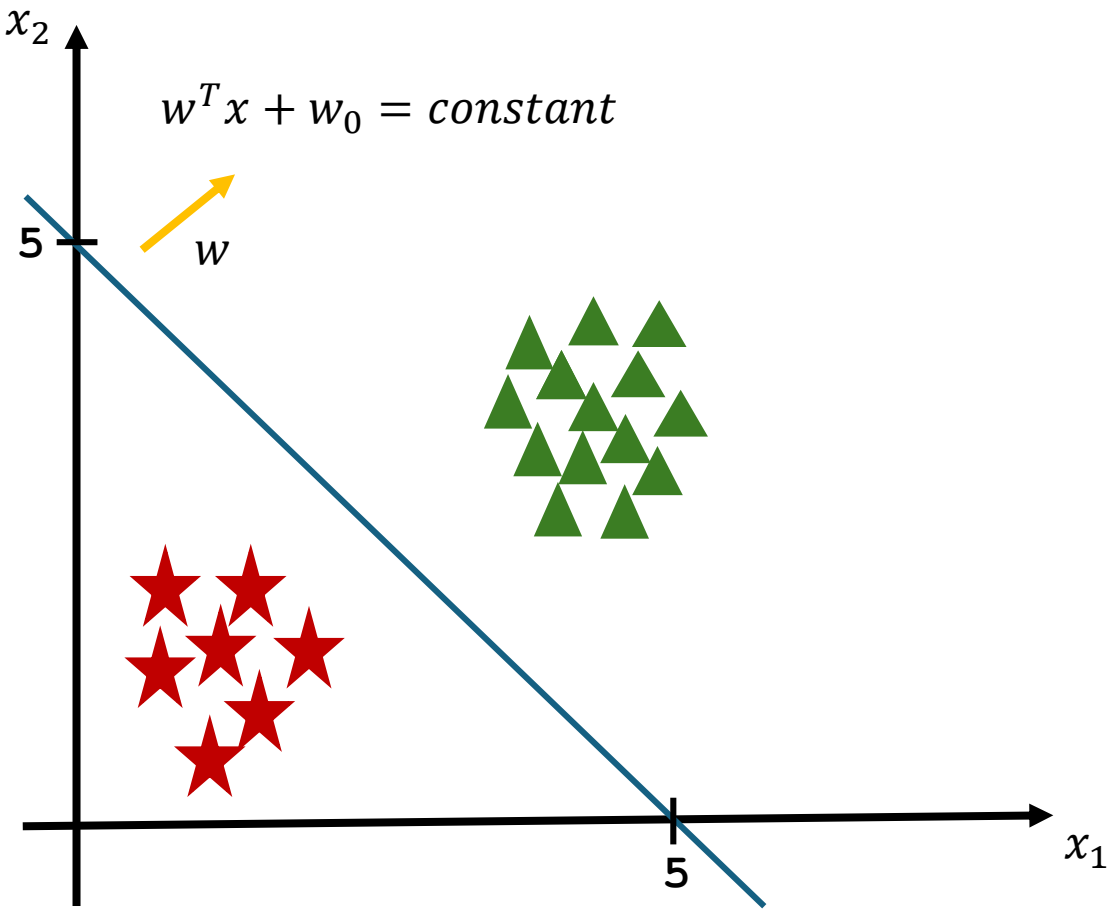
$f_{\overline{w},b}(\overline{x}) > 0.5$ class1

$f_{\overline{w},b}(\overline{x}) < 0.5$ class2

Decision Boundary



$w^T x + w_0 = constant$



Decision boundary

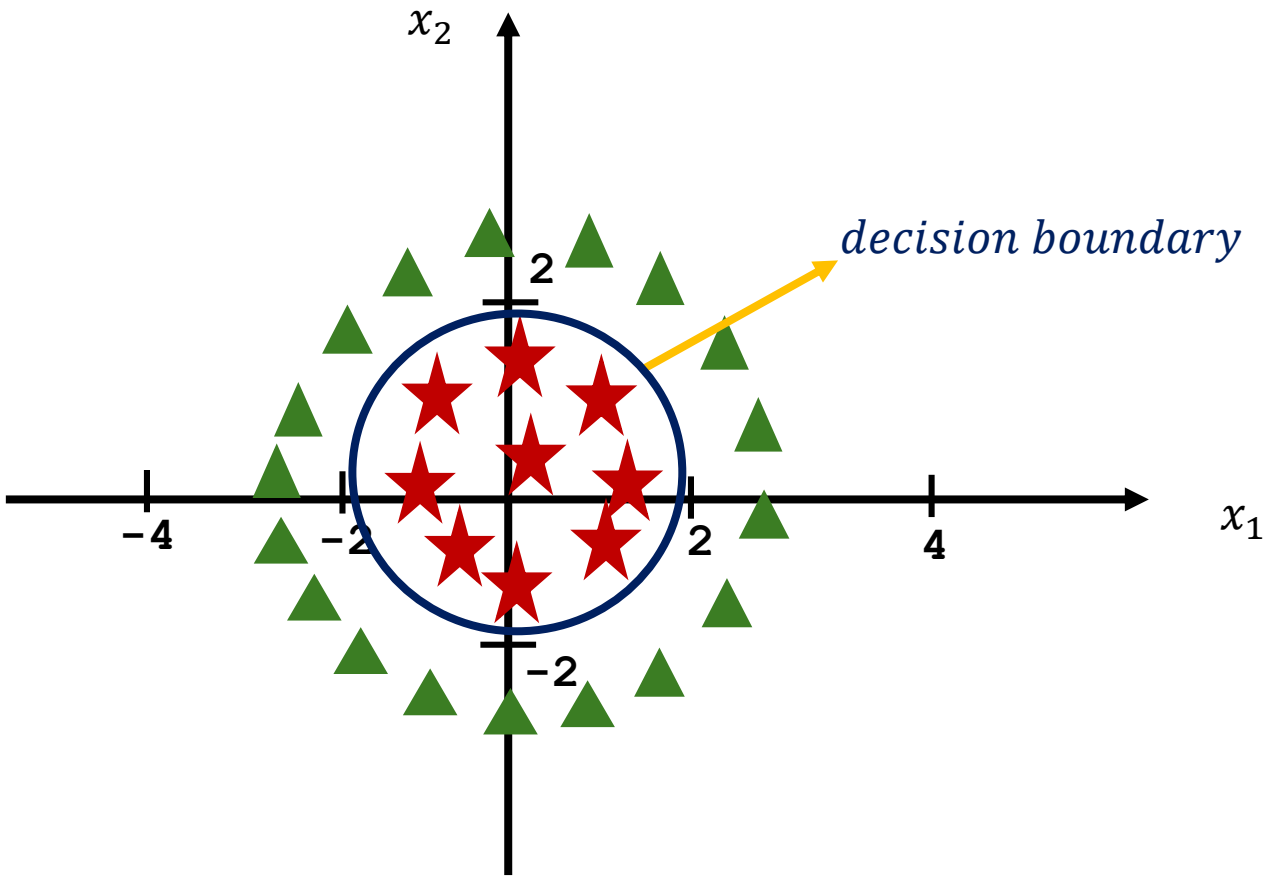
$$z = \overline{w}.\overline{x} + \mathrm{b} = 0$$

$$x_1 + x_2 = 5$$

$$f_{\overline{w},b}(\overline{x}) = g(z) = g(x_1 w_1 + x_2 w_2 + b)$$

1 1 −5

Non Linear Decision Boundary



Decision boundary

$$z = x_1^2 + x_2^2 - 2 = 0$$

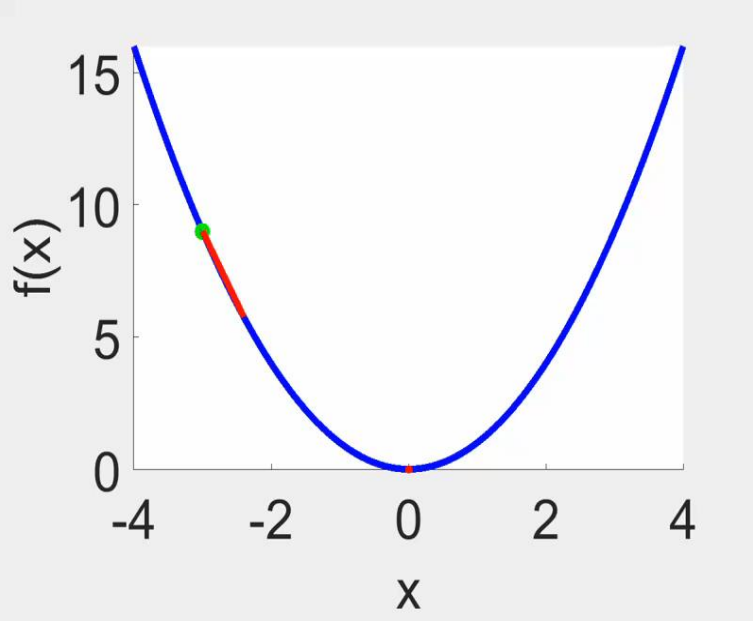
$$x_1^2 + x_2^2 = 2$$

$$f_{\overline{w},b}(\overline{x}) = g(z) = g(x_1^2 w_1 + x_2^2 w_2 + b)$$

1 1 -2

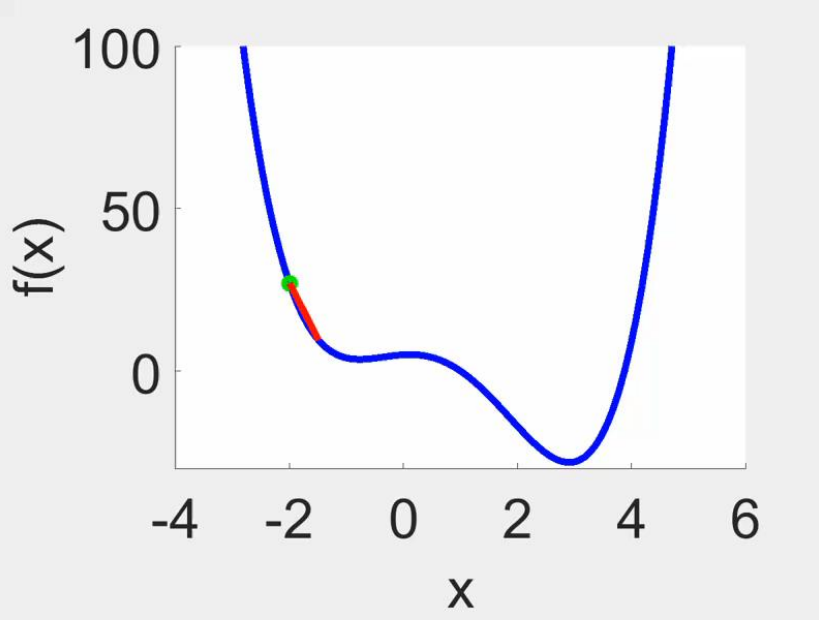
Squared Error Cost

$$J(\overline{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{2}\left(f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)}\right)^2$$



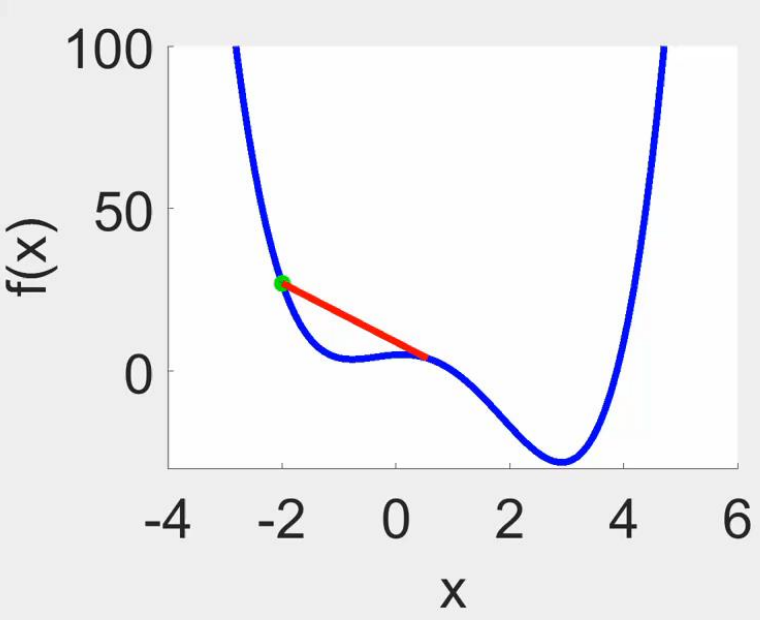
Convex

Linear regression



Non-Convex
Small learning rate

Logistic regression

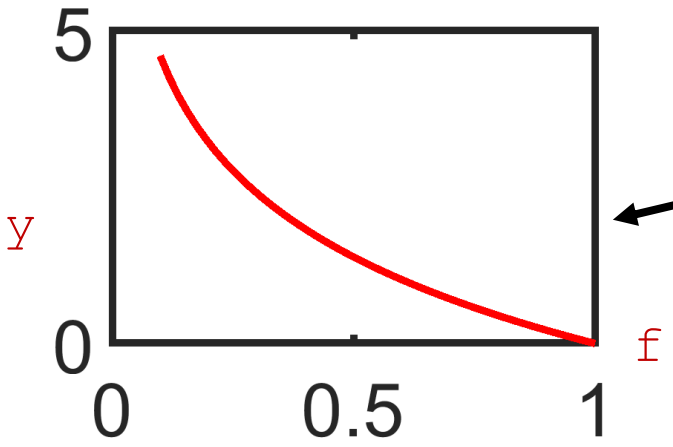


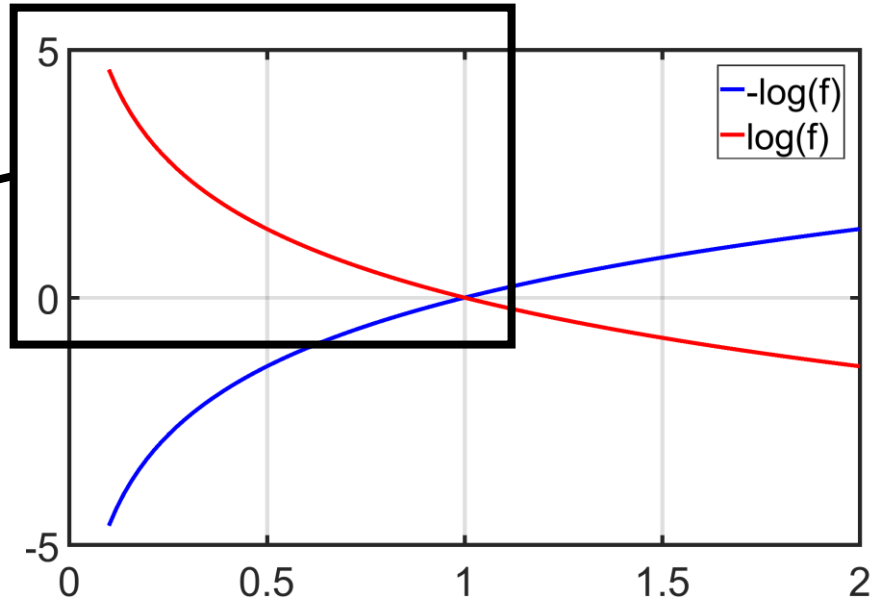
Non-Convex
Big learning rate

COST

$$J(\overline{w}, b) = \frac{1}{m}\sum_{i=1}^{m} L(f_{\overline{w},b}(x^{(i)}), y^{(i)})$$

$$L(f_{\overline{w},b}(x^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\overline{w},b}(x^{(i)}) & if\ y^{(i)} = 1 \\ -\log\left(1 - f_{\overline{w},b}(x^{(i)})\right) & if\ y^{(i)} = 0 \end{cases}$$

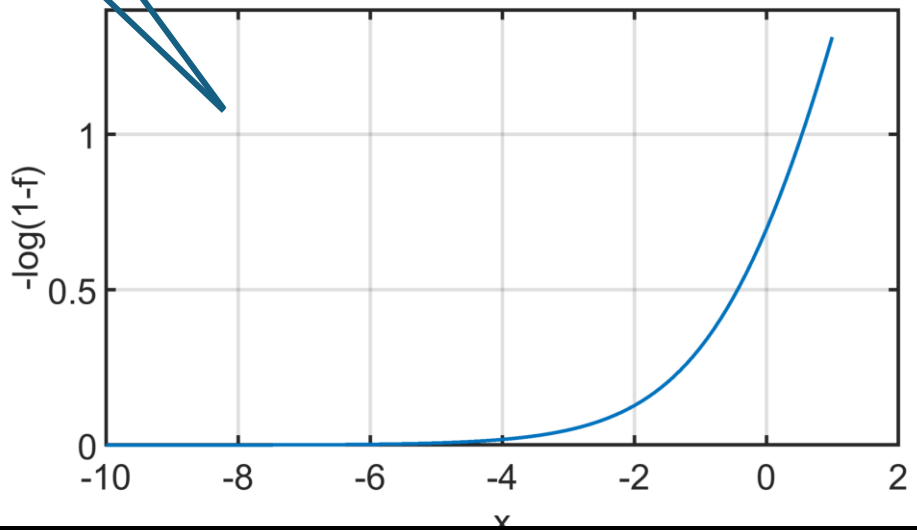




COST

$$J(\overline{w}, b) = \frac{1}{m}\sum_{i=1}^{m} L(f_{\overline{w},b}(x^{(i)}), y^{(i)})$$

$$L(f_{\overline{w},b}(x^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{\overline{w},b}(x^{(i)}) & if\ y^{(i)} = 1 \\ -\log\left(1 - f_{\overline{w},b}(x^{(i)})\right) & if\ y^{(i)} = 0 \end{cases}$$



SIMPLIFIED COST FUNCTION

$$L\left(f_{\overline{w},b}\left(x^{(i)}\right), y^{(i)}\right) = -y^{(i)}\log\left(f_{\overline{w},b}\left(x^{(i)}\right)\right) - \left(1 - y^{(i)}\right)\log\left(1 - f_{\overline{w},b}\left(x^{(i)}\right)\right)$$

$$J(\overline{w}, b) = \frac{1}{m}\sum_{i=1}^{m} L(f_{\overline{w},b}\left(x^{(i)}\right), y^{(i)})$$

$$= -\frac{1}{m}\sum_{i=1}^{m}[\, y^{(i)}\log\left(f_{\overline{w},b}\left(x^{(i)}\right)\right) - \left(1 - y^{(i)}\right)\log\left(1 - f_{\overline{w},b}\left(x^{(i)}\right)\right)]$$

Gradient Descent Implementation

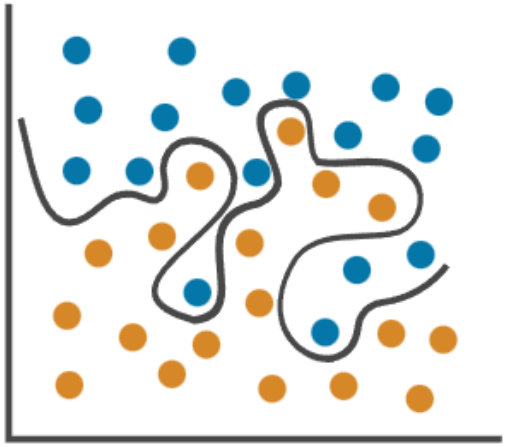
$$J(\overline{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}[\, y^{(i)}\log\left(f_{\overline{w},b}\left(x^{(i)}\right)\right) - \left(1 - y^{(i)}\right)\log\left(1 - f_{\overline{w},b}\left(x^{(i)}\right)\right)]$$

$$w_j = w_j \; -\propto \frac{\partial}{\partial w_j} \; J(\overline{w}, b) \longrightarrow \frac{\partial}{\partial w_j} \; J(\overline{w}, b) = \frac{1}{m}\sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)})x_j^{(i)}$$

$$b = b \; -\propto \frac{\partial}{\partial w_j} \; J(\overline{w}, b) \longrightarrow \frac{\partial}{\partial w_j} \; J(\overline{w}, b) = \frac{1}{m}\sum_{i=1}^{m} f_{\overline{w},b}\left(x^{(i)}\right) - y^{(i)})$$

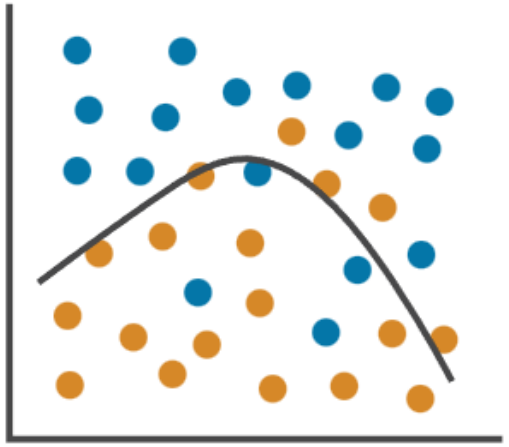
FITTING PROBLEMS

Overfitting



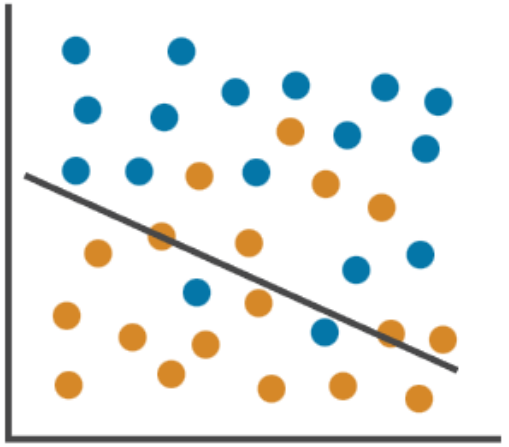
$\theta_0 + \theta_1 x$

Right Fit



$\theta_0 + \theta_1 x + \theta_2 x^2$

Underfitting

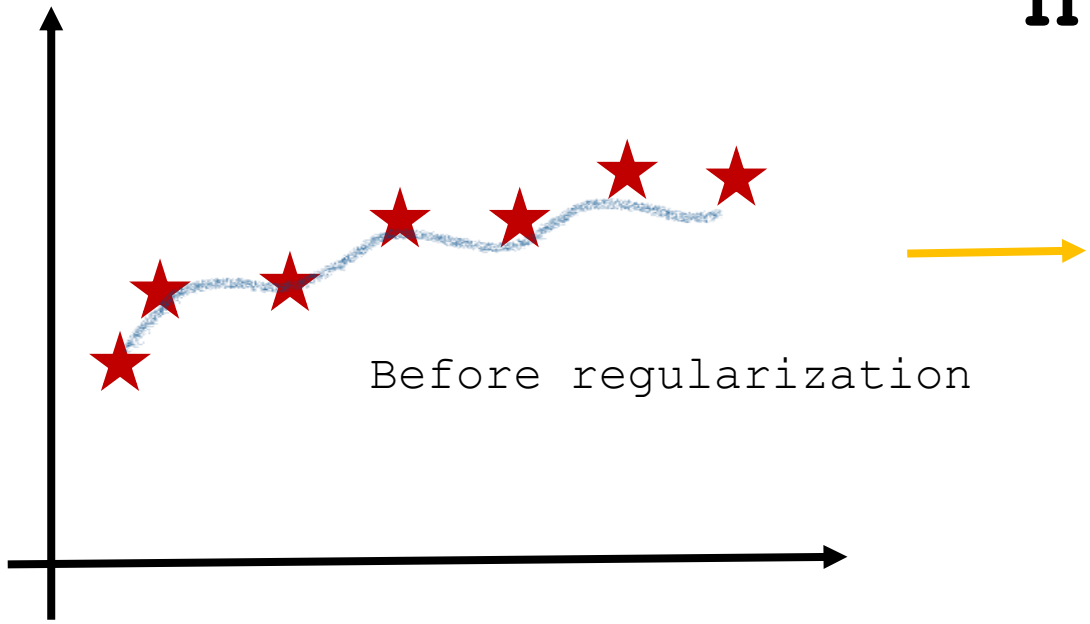


$\theta_0 + \theta_1 x + \theta_2 x^2 + \ \theta_2 x^3 + \theta_2 x^4 + \cdots$

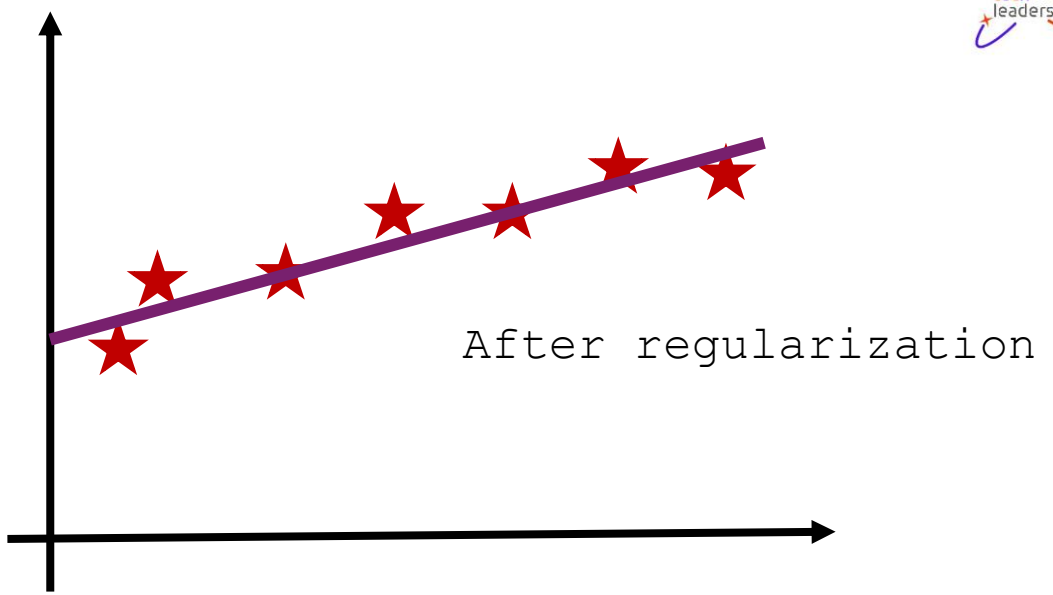
Error	Overfitting	Right Fit	Underfitting
Training	Low	Low	High
Test	High	Low	High

To reduce the impact of this tradeoff we use the concept of **Regularization**

Regularizatio
n



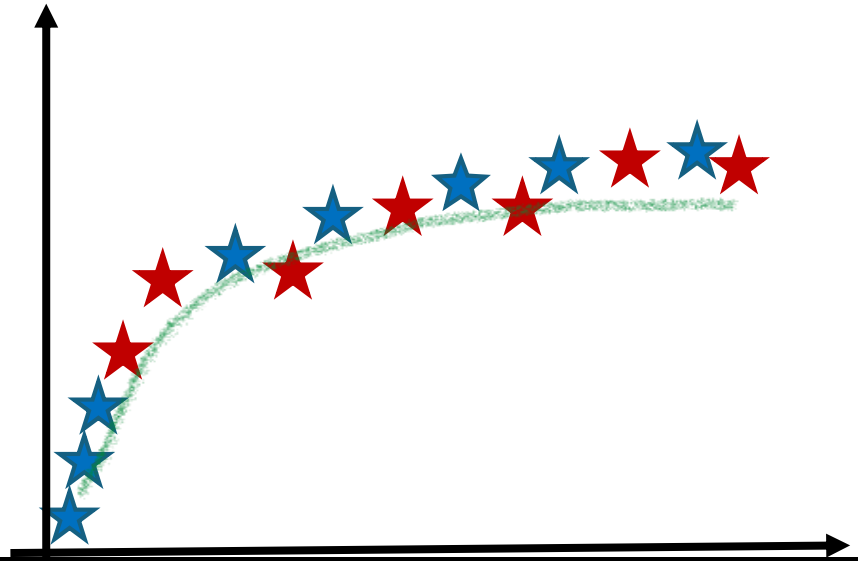
Before regularization



After regularization



After collect more data



Regularization

- We know, The General Equation of a Linear Regression can be expressed as follows:-
- $Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_N$

$$RSS = \sum_{i=1}^{m} (y_i - \beta_0 - \sum_{j=1}^{m} \beta_j X_{ij})^2$$

Evaluating classification

A confusion matrix compares the predicted classes with the actual classes.
Suppose a model is used to identify a defined agricultural stress event.
These are illustrative counts from 200 observations.

Four outcomes

- True positive (TP): Correctly identifies a positive event.
- False positive (FP): Flags an event that was not positive.
- False negative (FN): Fails to identify an actual positive event.
- True negative (TN): Correctly identifies a negative event.

In this example, the model correctly identifies 40 positive observations and misses 10.

	Actual positive	Actual negative
Predicted positive	TP = 40	FP = 15
Predicted negative	FN = 10	TN = 135

Accuracy, precision, recall and F1-score

Use the confusion matrix from the previous

Accuracy — overall proportion of correct predictions

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Accuracy = \frac{40 + 135}{200} = 87.5\%$$

Precision — how reliable are positive predictions?

$$Precision = \frac{TP}{TP + FP} = \frac{40}{55} \approx 72.7\%$$

Recall — how many actual positive cases were detected?

$$Recall = \frac{TP}{TP + FN} = \frac{40}{50} = 80\%$$

F1-score — balance between precision and recall

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}$$

For this example, $F1 \approx 76.2\%$

Interpretation

- Accuracy answers: What proportion of all predictions were correct?
- Precision answers: Of the predicted positive cases, what proportion were actually positive?
- Recall answers: Of the actual positive cases, what proportion did the model identify?
- F1-score balances precision and recall using their harmonic mean.